ConnectTel

Customer Churn Prediction

Machine Learning Internship Project

SN PRANAV | May 2026 | Telco Customer Churn Dataset

26.5% Churn Rate	**0.8470** XGBoost AUC-ROC	**79.7%** Churner Recall	**3,471%** Estimated ROI

1. PROBLEM STATEMENT

ConnectTel is losing millions in revenue every quarter due to customer churn. Acquiring a new customer costs 5 to 7 times more than retaining an existing one. This project builds a machine learning system that proactively identifies customers at high risk of churning, enabling the retention team to intervene with targeted offers.

Key business metrics:

5–7×	Cost to acquire vs retain a customer
26.5%	Quarterly churn rate at ConnectTel
$120	Average monthly revenue lost per churner
$25	Cost of one retention offer

2. EDA KEY FINDINGS

Dataset: 7,043 customers × 21 features. TotalCharges had 11 blank string entries for new customers with tenure = 0, imputed with 0.

Class Imbalance: 26.5% churn rate (1,869 churners vs 5,174 loyal). AUC-ROC used as primary metric — accuracy is misleading here.

Hypothesis 1: Fiber Optic users churn at 41.9% vs DSL at 18.9% ($p < 0.05$, statistically significant). Difference = 23 percentage points.

Hypothesis 2: Customers without a partner churn at 32.9% vs 19.7% with a partner — family ties are a strong retention signal.

Contract Type: Month-to-month customers in their first 12 months have a 50%+ churn rate — the highest risk group in the dataset.

Service Count: Customers with 4+ add-on services churn at half the rate of single-service customers — engagement = retention.

6 Features Engineered:

Feature	Formula	Business Rationale
TotalChargesPerTenure	TotalCharges / tenure	Average monthly spend proxy
ServiceCount	Sum of 8 add-on flags	Engagement signal (0–8)
HasFamilyTies	Partner OR Dependents	Stability — family customers churn less
IsNewCustomer	tenure <= 12 months	Flags the most vulnerable window
IsHighValue	Charges > 75th pct	High price + M2M = highest risk
ContractRisk	M2M=2, 1yr=1, 2yr=0	Ordinal risk encoding

3. MODEL RESULTS

Three models were trained with StratifiedKFold(n_splits=5) cross-validation. class_weight='balanced' and scale_pos_weight=2.77 handle the 74/26 imbalance. All models evaluated on the same held-out 20% test set (random_state=42).

Model	AUC-ROC	Recall	Precision	F1	Accuracy
Logistic Regression	0.8410	0.7914	0.4992	0.6122	0.7339
Random Forest	0.8442	0.7380	0.5691	0.6426	0.7821
XGBoost (tuned)	**0.8470**	**0.7968**	**0.5129**	**0.6241**	**0.7452**

XGBoost achieved the highest AUC-ROC (**0.8470**) and highest Recall (**79.7%**), correctly identifying 79.7% of all actual churners. GridSearchCV searched 24 combinations × 5 folds = 120 fits. Best params: n_estimators=300, max_depth=5, learning_rate=0.05, subsample=0.8.

Optimal decision threshold: 0.37 (not default 0.50) — maximises revenue saved at $120 LTV vs $25 offer cost per customer.

4. SHAP INTERPRETABILITY

SHAP (SHapley Additive exPlanations) converts XGBoost's internal calculations into plain feature contributions. Positive SHAP = pushes toward churn. Negative SHAP = pushes toward staying. Three plot types were generated:

Summary (Beeswarm)	Global view — all features ranked by mean |SHAP|. Each dot = one customer.
Dependence Plots	How one feature's SHAP changes as its value increases. Covers MonthlyCharges, Tenure, and ContractF
Waterfall Plots	Individual explanation for 3 specific customers: high-risk churner, safe stayer, and borderline case.

Top 5 Churn Drivers (by mean |SHAP|):

Rank	Feature	Mean |SHAP|	Direction	Interpretation
#1	ContractRisk	0.692	Positive	Month-to-month = highest churn driver
#2	Tenure	0.461	Negative	Longer tenure = strong protection
#3	InternetService_Fiber	0.310	Positive	Fiber users consistently high risk
#4	MonthlyCharges	0.249	Positive >$65	Charges above $65 turn sharply positive
#5	ServiceCount	0.141	Negative	More services = lower churn risk

5. BUSINESS RECOMMENDATIONS

01

Target Month-to-Month Customers in Year 1

Segment: tenure < 12 months + Month-to-month contract
Action: 15% discount to upgrade to 1-year contract
Impact: ~$5,030/month revenue protected

02

Fiber Optic Bundle Offer

Segment: Fiber Optic + MonthlyCharges > $65
Action: Free OnlineSecurity + TechSupport for 12 months
Impact: ~$3,930/month revenue protected

03

Onboarding Excellence Programme

Segment: All customers with tenure <= 12 months
Action: Structured check-ins at months 1, 3, 6, 9 + loyalty reward
Impact: ~$9,432/month revenue protected

04

Service Upsell Campaign

Segment: ServiceCount <= 2 + Month-to-month
Action: First 3 months free on 2 additional add-ons
Impact: Reduces churn rate by approximately 15%

05

Use Threshold 0.37 in Production

Segment: All customers with predicted churn probability > 0.37
Action: Flag 633 customers for outreach vs 298 at default 0.50
Impact: Catches 392 true churners — estimated ROI of 3,471%

6. COST-BENEFIT SUMMARY

Metric	Value	Basis
Customers flagged (thr=0.37)	633	Predicted churn prob > 0.37
True positives (real churners)	~392	62% of flagged are real churners
Campaign cost	$15,825	633 customers × $25/offer
Revenue protected (monthly)	$47,040	392 × $120 avg monthly LTV
Revenue protected (12 months)	$564,480	Annual projection
Estimated ROI	3,471%	(Revenue - Cost) / Cost × 100

7. LIMITATIONS & NEXT STEPS

Correlation vs Causation: SHAP shows which features the model uses, not why customers churn. Fiber Optic users may churn due to price, competition, or quality — the model cannot distinguish these.

Offer Acceptance: Revenue estimates assume 20–25% acceptance rate. A/B testing is required to validate the campaign's true effectiveness before scaling.

Model Drift: Customer behaviour changes quarterly. The model should be retrained on fresh data every 3 months.

Next Step — Dashboard: Build a Streamlit web app where the retention team can input customer details and receive instant churn probability + SHAP explanation.

Next Step — Deployment: Wrap the model in a REST API (Flask/FastAPI) and integrate with ConnectTel's CRM system for automated flagging.
